

Understanding Dataset & Labeling

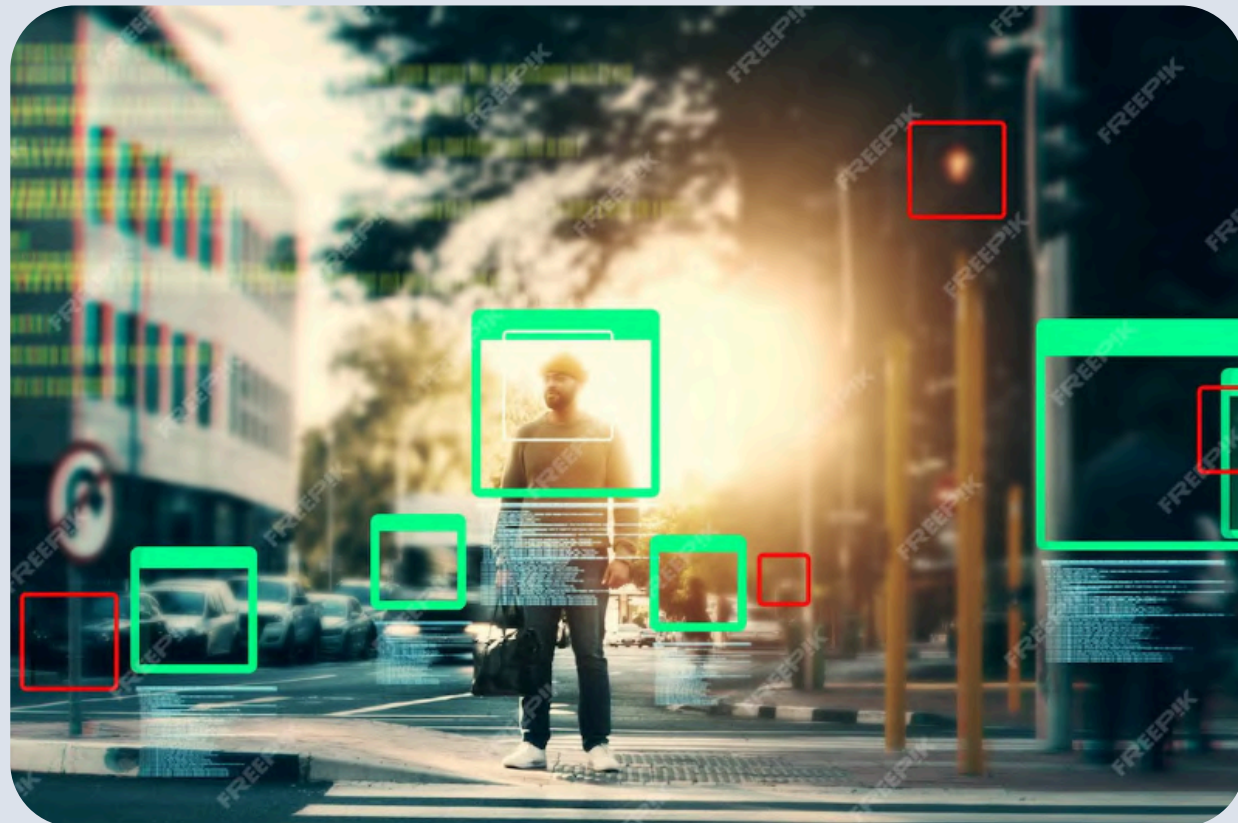
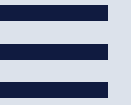
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WHERE INNOVATION
MEETS OPPORTUNITY.



Artificial Intelligence (AI) is no longer just a futuristic idea—it's a powerful tool shaping businesses today. From customer service to logistics, AI is driving efficiency, innovation, and new market opportunities across industries.



What is Dataset

A dataset is a structured collection of data, often organized in a table, spreadsheet, or database format. It brings together related information that has been collected for a specific purpose, such as analysis, research, or to train machine learning models.

Organization

- STRUCTURING DATA
- FOLDER STRUCTURE
- CSV (COLUMN, ROW)

Purpose

DATASETS ARE
USUALLY CREATED
WITH A SPECIFIC
GOAL IN MIND

Variety of Data Types

INCLUDING NUMERICAL
VALUES, TEXT, IMAGES,
AUDIO RECORDINGS,
AND MORE

Essential for Data- Driven Activities

DATASETS POWER
DATA, AI, AND ML.



Data Collection Strategy

SURVEYS AND
QUESTIONNAIRES

INTERVIEWS

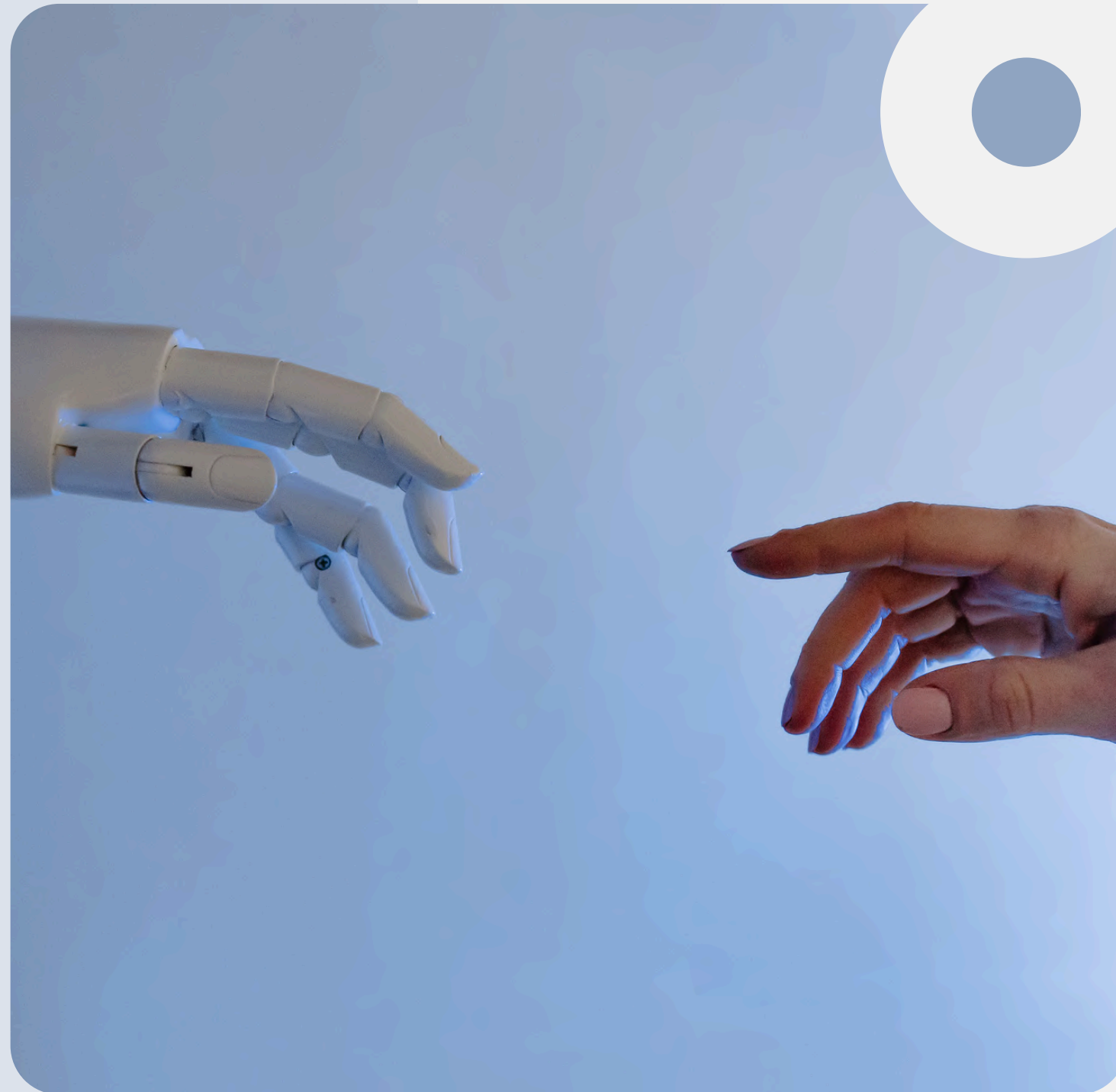
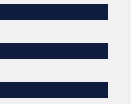
FOCUS GROUPS

OBSERVATIONS

EXPERIMENTS

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Choosing the Right Strategy

PERSONALIZATION POWERED BY INTELLIGENCE.

Businesses are using AI to create more personalized, responsive, and engaging customer experiences. Chatbots, virtual assistants, and recommendation engines are helping brands stay connected to customers 24/7, enhancing loyalty and boosting sales.



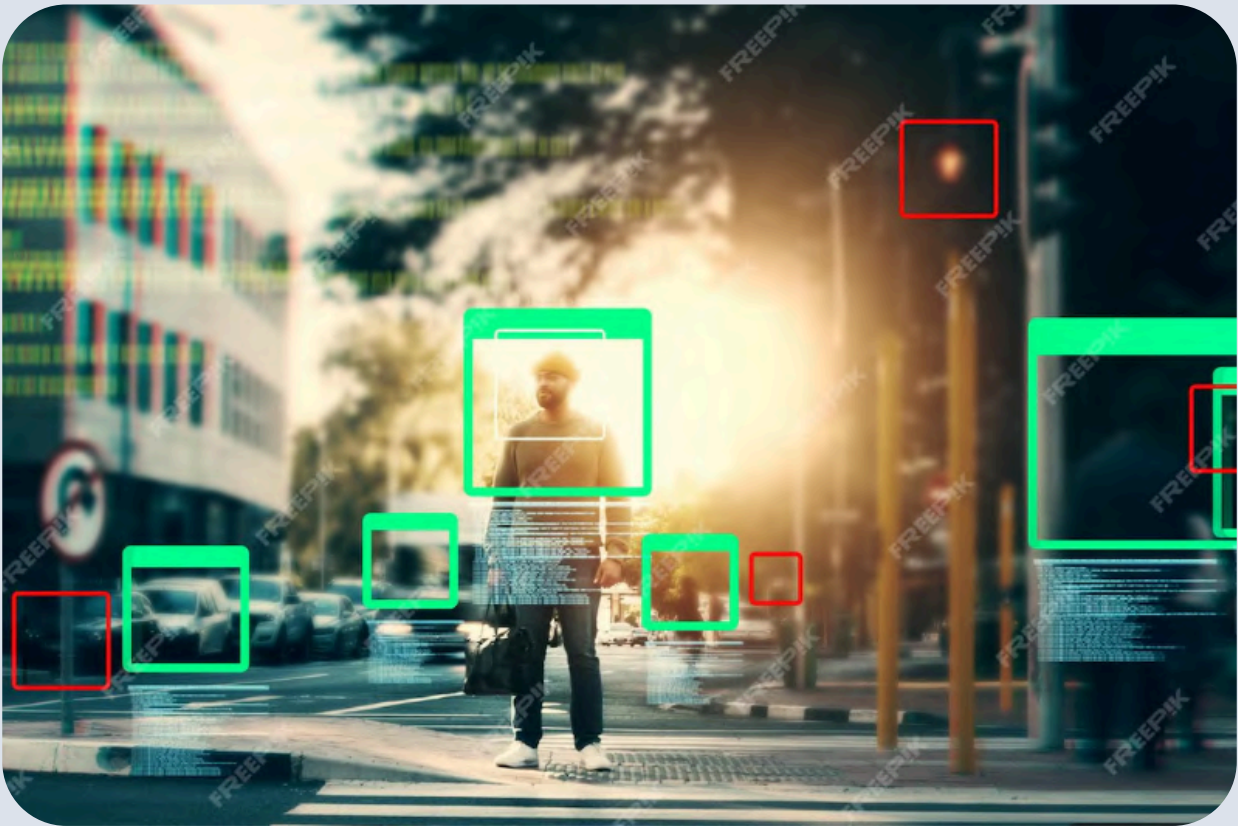
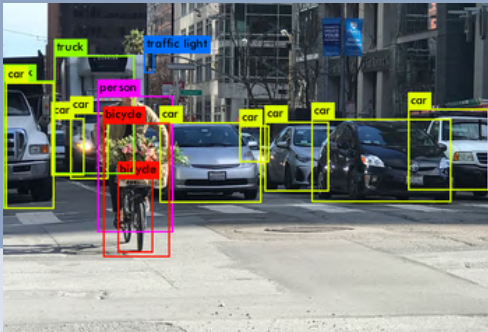


Image Labeling Techniques

Classification Labeling



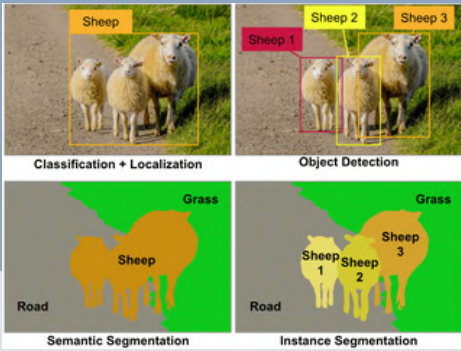
Object Detection Labeling



Semantic Segmentation



Instance Segmentation



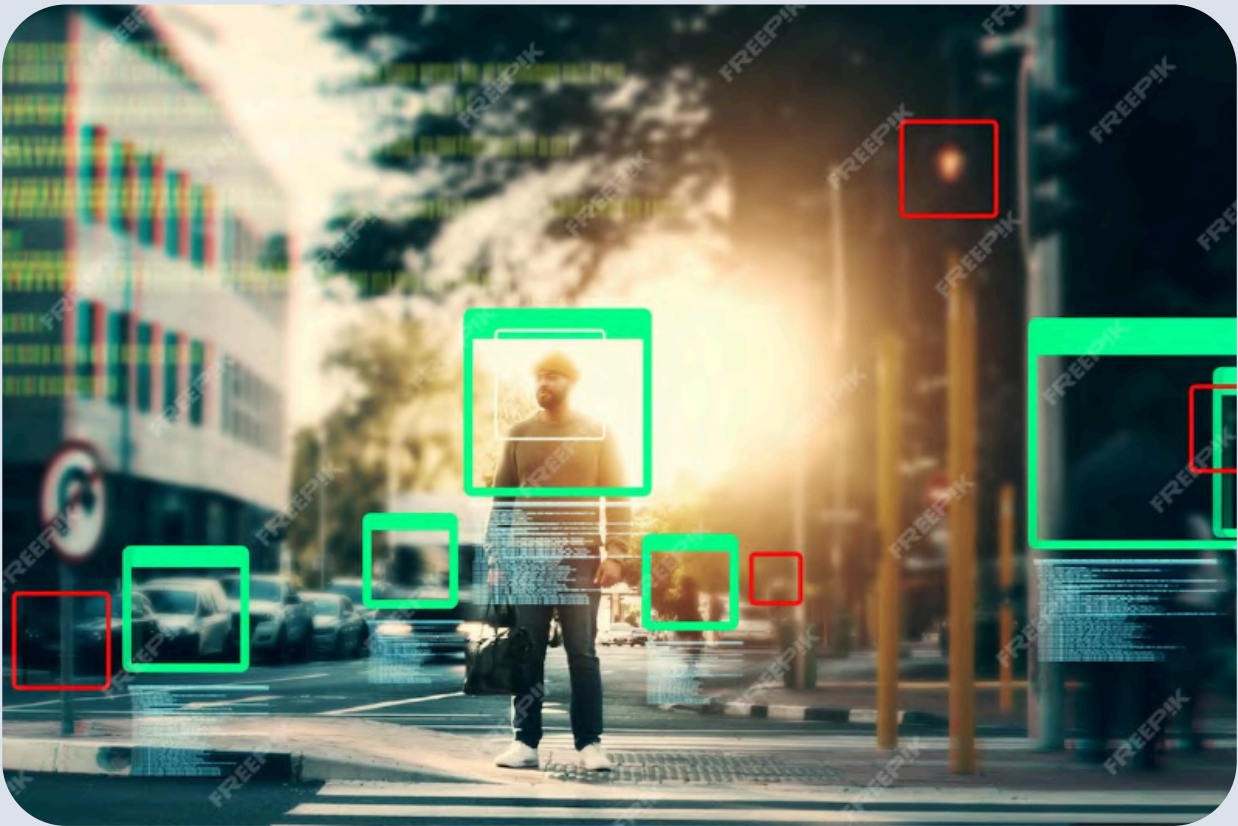
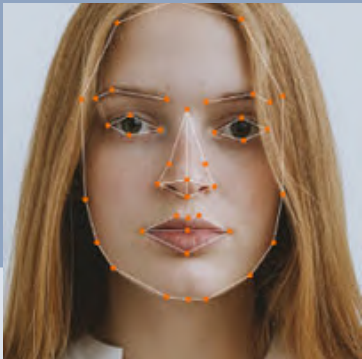


Image Labeling Techniques

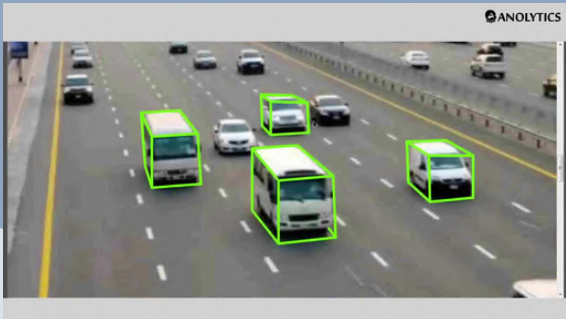
Polygonal Segmentation



Keypoint/Landmar k Annotation



3D Cuboid Annotation

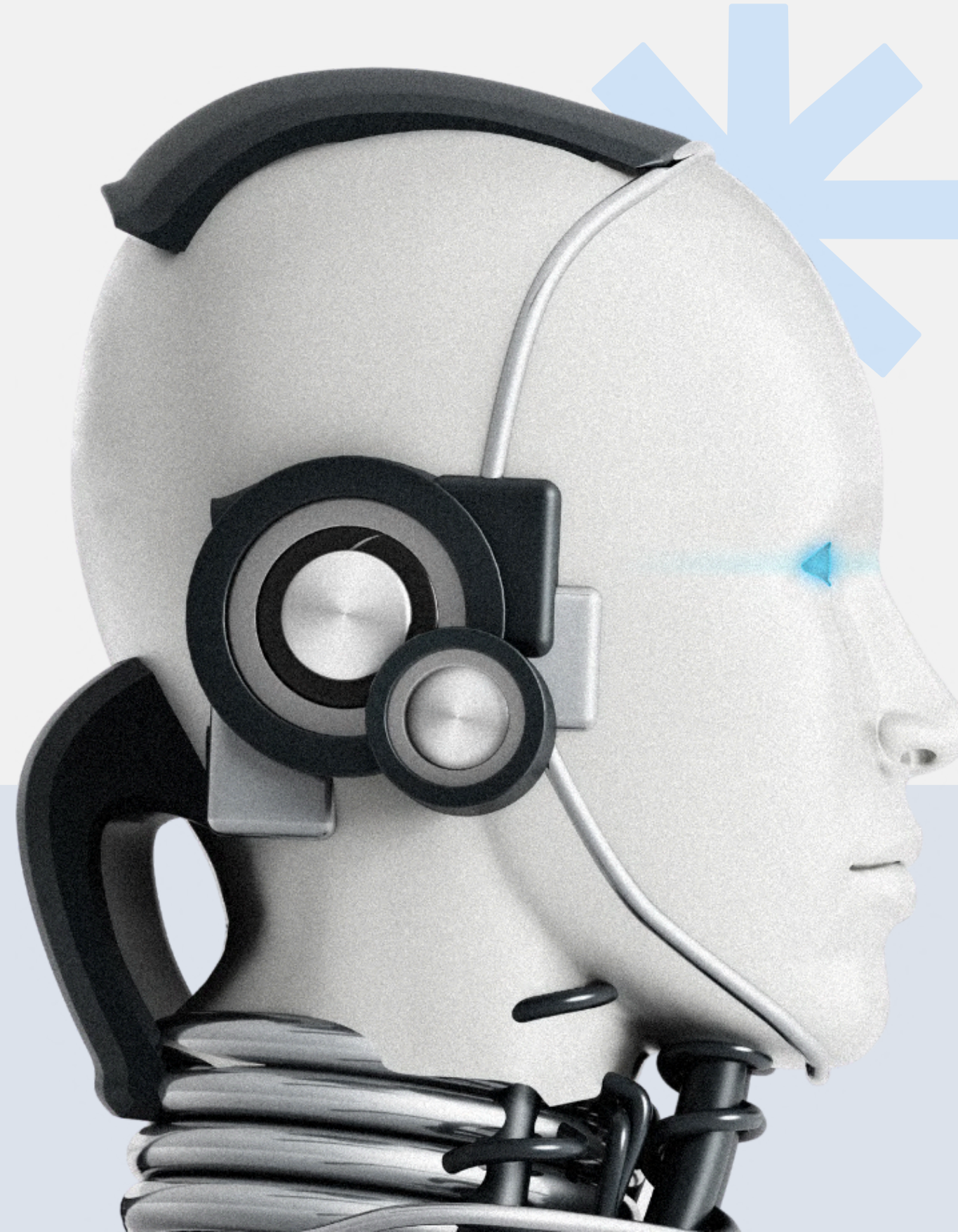




Data Labeling Tool and Format

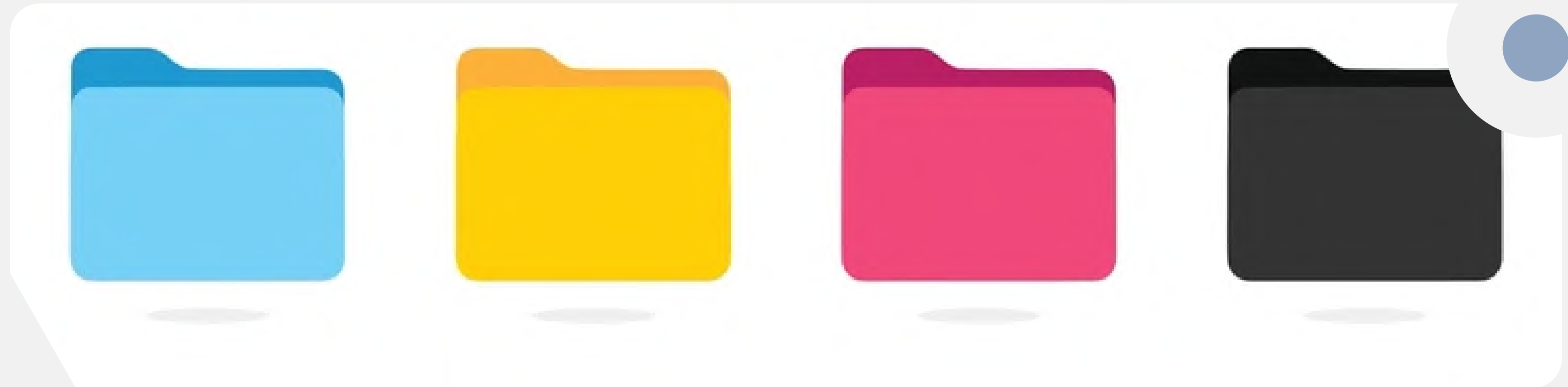
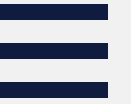
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DATA
DECISIONS—AND
DESTINY. DRIVES



A data labeling tool is software used to annotate raw data (like images, text, or audio) with meaningful tags or labels. This process transforms unstructured data into structured datasets, which are essential for training machine learning models. The labeling format refers to the standardized way these annotations are stored, ensuring consistency and compatibility with AI development frameworks (e.g., JSON, XML, or specific schemas like COCO or YOLO).





Labeling Classification

- Folder name is the Label
- Images inside folder is data to be processed and trained

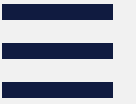




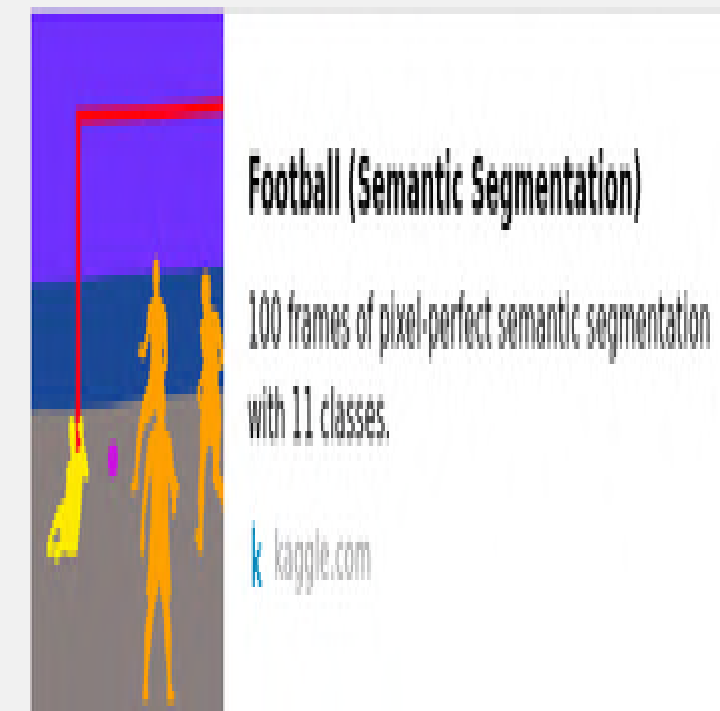
Labeling Object Detection

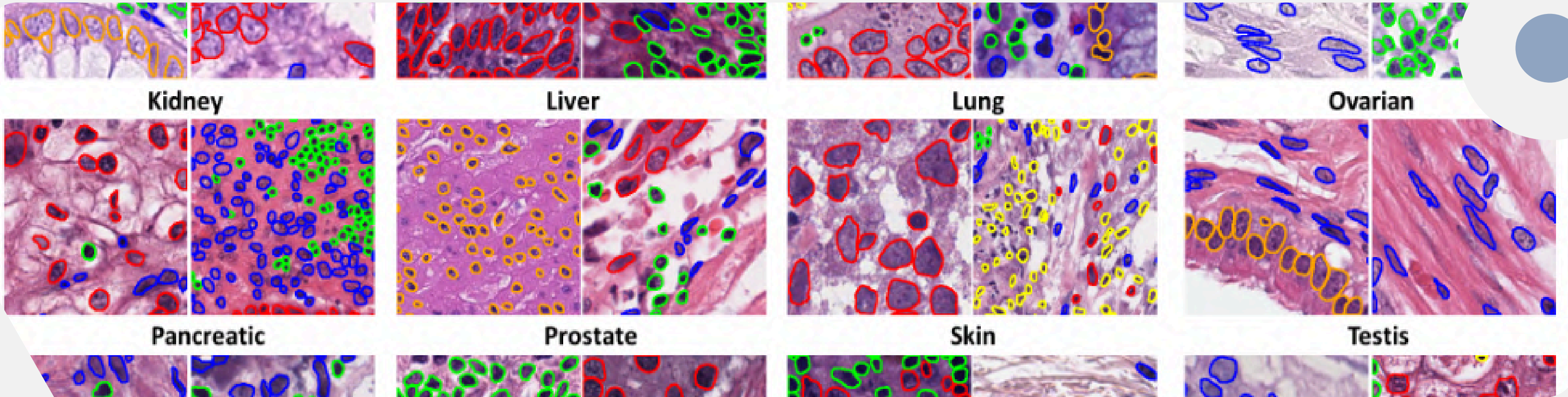
- Bounding box with label in json or XML



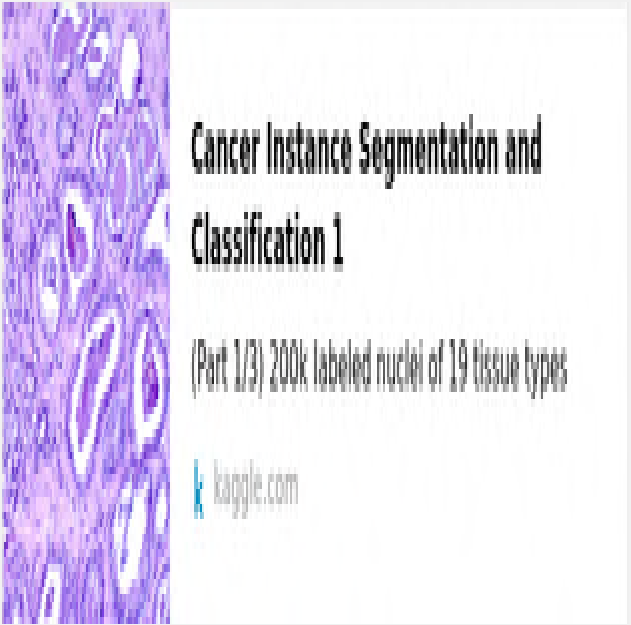


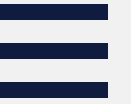
Labeling Semantic Segmentation





Labeling Instance Segmentation

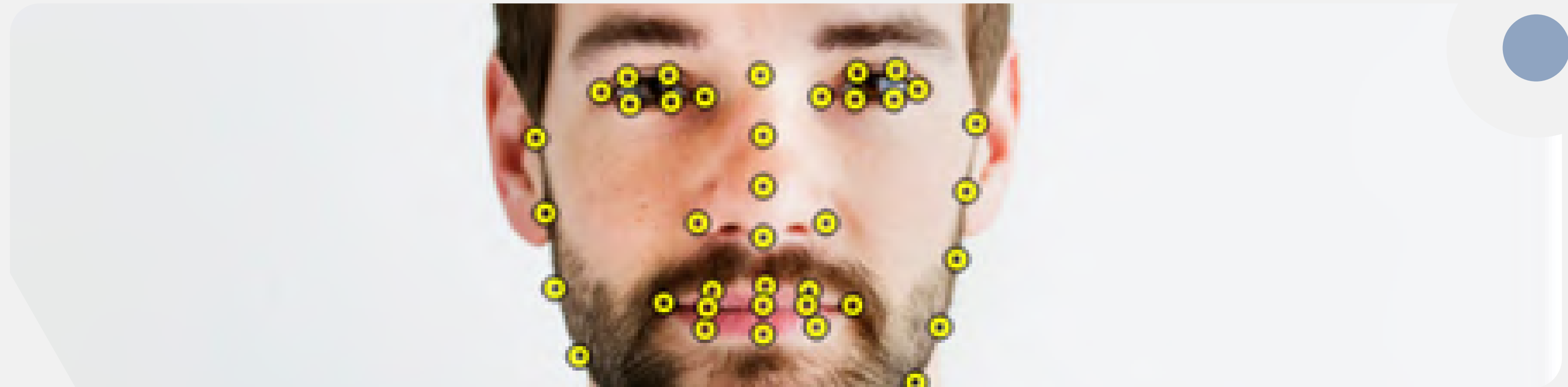
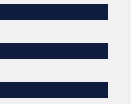




Labeling

Polygonal Segmentation

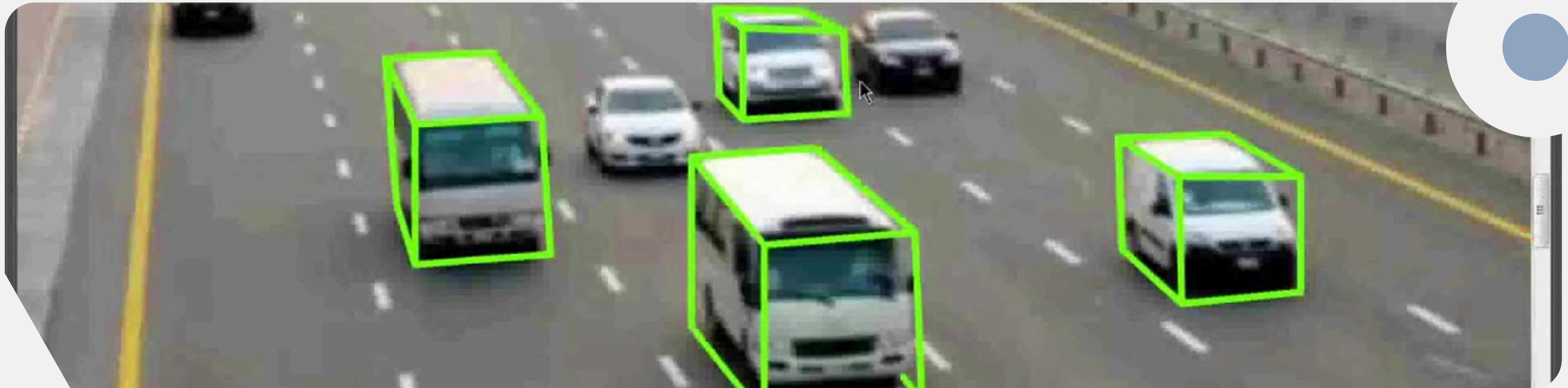
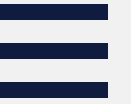




Labeling Keypoint/Landmark Annotation

<https://www.kaggle.com/datasets/unidpro/facial-keypoint-detection>

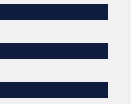




Labeling 3D Cuboid Annotation



<https://www.kaggle.com/datasets/unidpro/facial-keypoint-detection>



Data Collection Quality

ACCURACY

COMPLETENESS

CONSISTENCY

TIMELINESS/
FRESHNESS

UNIQUENESS

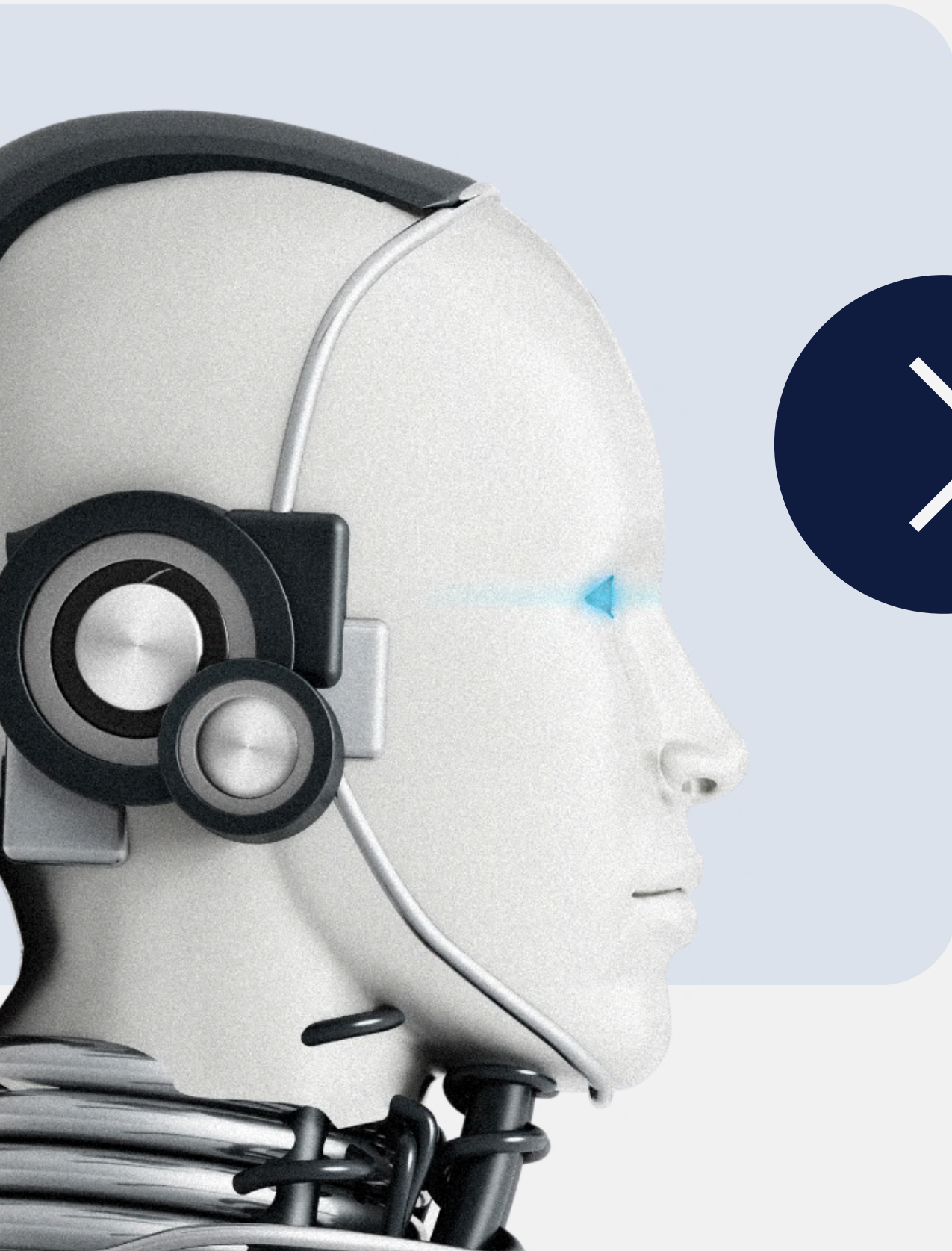
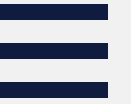
VALIDITY

RELEVANCE/
FITNESS FOR
PURPOSE



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Laneline detection Strategy

TOMORROW'S BREAKTHROUGHS START TODAY.

How about your idea?

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Thank you



For Joining